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Ingot puller apparatus having a flange that extends from the funnel or from the silicon feed tube

Abstract

Methods for growing single crystal silicon ingots that involve silicon feed tube inert gas control are disclosed. Ingot puller apparatus that include a flange that extends radially from a silicon funnel or from a silicon feed tube to reduce backflow of gases from the silicon feed tube into the growth chamber are also disclosed.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. Non-Provisional patent application Ser. No. 17/570,146, filed Jan. 6, 2022, which is incorporated herein by reference in its entirety.

FIELD

(1) The field of the disclosure relates to methods for growing single crystal silicon ingots that involve silicon feed tube inert gas control and to ingot puller apparatus that include a flange that extends radially from a silicon funnel or from a silicon feed tube to reduce backflow of gases from the silicon feed tube into the growth chamber.

BACKGROUND

(2) Continuous Czochralski (CCz) is well suited to form 300 mm or 200 mm diameter single crystal silicon ingots. Continuous Czochralski methods involve forming a single crystal silicon ingot from a melt of silicon while continuously or intermittently adding solid polycrystalline silicon to the melt to replenish the melt while the ingot is grown. The methods may involve forming multiple ingots from the same melt while the hot zone remains at temperature (i.e., with a melt continuously being present in the crucible assembly while the plurality of ingots is grown).

(3) Solid-state silicon is added to the melt through a silicon feed tube that directs the silicon to the melt. Periodically, the silicon feed tube becomes occluded and silicon cannot pass through the tube. Clogging of the feed tube causes the ingot production run to be stopped and the ingot puller apparatus to be taken out of service which reduces throughput and increases cost. An inert gas such as argon may be passed through the tube to reduce clogging. However, the flow of inert gas can cause inert gas bubbles (e.g., argon bubbles) to form in the melt which impacts the void count in the resulting ingot and sliced wafers. Customers expect that wafers grown by continuous Czochralski have the same relatively low void count as wafers grown by standard batch Czochralski.

(4) A need exists for ingot puller apparatus and methods for forming an ingot that reduce occlusion of the silicon feed tube and that do not result in formation of excessive inert gas bubbles in the melt.

(5) This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the disclosure, which are described and/or claimed below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it should be understood that these statements are to be read in this light, and not as admissions of prior art.

SUMMARY

(6) One aspect of the present disclosure is directed to a method for growing a single crystal silicon ingot in a continuous Czochralski process in an ingot puller apparatus. The ingot puller apparatus includes a housing defining a growth chamber, a crucible assembly disposed in the growth chamber, and a silicon feed tube for adding solid silicon to the crucible assembly. A melt of silicon is formed in a crucible assembly. Heat is applied to the melt in a stabilization phase. An inert gas is introduced into the silicon feed tube at a first inert gas feed rate during the stabilization phase. A surface of the melt is contacted with a seed crystal. A single crystal silicon ingot is withdrawn from the melt in an ingot growth phase. The inert gas is introduced into the silicon feed tube at a second inert gas feed rate during the ingot growth phase. The first inert gas feed rate is greater than the second inert gas feed rate.

(7) Another aspect of the present disclosure is directed to a method for growing a single crystal silicon ingot in a continuous Czochralski process in an ingot puller apparatus. The ingot puller apparatus includes a housing defining a growth chamber, a crucible assembly disposed in the

growth chamber, and a silicon feed tube for adding solid silicon to the crucible assembly. A melt of silicon is formed in a crucible assembly. Heat is applied to the melt in a stabilization phase. An inert gas is introduced into the silicon feed tube at a first inert gas feed rate during the stabilization phase. The first inert gas feed rate is between 2.5 and 3.5 slpm. A surface of the melt is contacted with a seed crystal. A single crystal silicon ingot is withdrawn from the melt in an ingot growth phase. The inert gas is introduced into the silicon feed tube at a second inert gas feed rate during the ingot growth phase. The second inert gas feed rate is between 2.5 and 3.5 slpm.

(8) A further aspect of the present disclosure is directed to an ingot puller apparatus for manufacturing a single crystal silicon ingot. The ingot puller apparatus includes a crucible assembly for holding a silicon melt. A crystal puller housing defines a growth chamber for pulling a silicon ingot from the melt. The crucible assembly is disposed within the growth chamber. The ingot puller apparatus includes a silicon feed tube for adding solid silicon to the crucible assembly. A funnel at least partially extends into the silicon feed tube. The funnel and silicon feed tube are separated by a gap. A flange extends radially from the funnel or from the silicon feed tube. The flange extends across at least a portion of the gap to reduce backflow of gases from the silicon feed tube into the growth chamber.

(9) Various refinements exist of the features noted in relation to the above-mentioned aspects of the present disclosure. Further features may also be incorporated in the above-mentioned aspects of the present disclosure as well. These refinements and additional features may exist individually or in any combination. For instance, various features discussed below in relation to any of the illustrated embodiments of the present disclosure may be incorporated into any of the above-described aspects of the present disclosure, alone or in any combination.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a cross-section of an example ingot puller apparatus with a silicon charge disposed in the crucible assembly;
- (2) FIG. 2 is a cross-section of the ingot puller apparatus after melt-down and during the stabilization phase;
- (3) FIG. 3 is a cross-section of the ingot puller apparatus with the silicon seed crystal lowered to contact the melt to begin the ingot growth phase;
- (4) FIG. 4 is a cross-section of the ingot puller apparatus during the ingot growth phase;
- (5) FIG. 5 is a perspective view of a silicon feed system for adding silicon to the crucible assembly;
- (6) FIG. 6 is a detailed cross-section of the ingot puller apparatus showing the silicon feed tube;
- (7) FIG. 7 is a detailed cross-section of an embodiment of an ingot puller apparatus showing a flange that extends from the funnel;
- (8) FIG. 8 is a detailed cross-section of another embodiment of an ingot puller apparatus showing a flange that extends from the lower end of the funnel;
- (9) FIG. 9 is a simulation of the mass fraction of SiO in the silicon feed tube during the ingot growth phase;
- (10) FIG. 10 is a simulation of the mass fraction of SiO in the silicon feed tube during the stabilization phase; and
- (11) FIG. 11 shows box plots of the defect counts for wafers grown with an argon gas flow rate through the silicon feed tube of 12 slpm (Crystal 1 and Crystal 2) and for wafers grown with an argon gas flow rate through the silicon feed tube of 3 slpm (Crystal 3).

(12) Corresponding reference characters indicate corresponding parts throughout the drawings.

DETAILED DESCRIPTION

(13) Provisions of the present disclosure relate to methods for growing a single crystal silicon ingot

in a continuous Czochralski process in an ingot puller apparatus and to ingot puller apparatus having a flange that extends across a portion of a gap between the funnel and silicon feed tube. The methods and apparatus may reduce downstream clogging of the silicon feed tube.

(14) An example ingot puller apparatus (or more simply “ingot puller”) is indicated generally at **100** in FIG. 1. The ingot puller apparatus **100** includes a crucible assembly **102** for holding a melt **104** (FIG. 2) of semiconductor or solar-grade material silicon. The crucible assembly **102** is supported by a susceptor **106**. The ingot puller apparatus **100** includes a crystal puller housing **108** that defines a growth chamber **152** for pulling a silicon ingot **113** (FIG. 4) from the melt **104** along a pull axis A.

(15) The crucible assembly **102** has a sidewall **131** (FIG. 1) and one or more fluid barriers **121**, **130** or “weirs” that separate the melt into different melt zones. In the illustrated embodiment, the crucible assembly **102** includes a first weir **121**. The first weir **121** and sidewall **131** define an outer melt zone **143** (FIG. 2) of the silicon melt **104** and the crucible assembly **102**. The crucible assembly **102** includes a second weir **130** radially inward to the first weir **121** which defines an inner melt zone **127** of the silicon melt **104** and crucible assembly **102**. The inner melt zone **127** is the growth region from which the single crystal silicon ingot **113** (FIG. 4) is grown. The first weir **121** and the second weir **130** define a middle melt zone **132** of the crucible assembly **102** and silicon melt **104** in which the melt **104** may stabilize as it moves toward the inner melt zone **127**. The first and second weirs **121**, **130** each have at least one opening formed therein to permit molten silicon to flow radially inward towards the inner melt zone **127**.

(16) In the illustrated embodiment, the first weir **121**, second weir **130**, and sidewall **131** each have a generally annular shape. The first weir **121**, second weir **130**, and sidewall **131** may be part of three nested crucibles which are joined at the bottom or floor **129** of the crucible assembly **102** (i.e., the first and second weirs **121**, **130** are the sidewalls of two crucibles nested within a larger crucible). The crucible assembly configuration depicted in FIGS. 1-4 is exemplary. In other embodiments, the crucible assembly **102** has a single layer floor (i.e., does not have nested crucibles) with the weirs extending upward from the floor **129**. Optionally, the floor **129** may be flat rather than curved and/or the weirs **121**, **130** and/or sidewall **131** may be straight-sided. Further, while the illustrated crucible assembly **102** is shown with two weirs, in other embodiments the crucible assembly may have a single weir or even no weirs.

(17) The susceptor **106** is supported by a shaft **105**. The susceptor **106**, crucible assembly **102**, shaft **105**, and ingot **113** (FIG. 4) have a common longitudinal axis or “pull axis” A.

(18) A pulling mechanism **114** is provided within the ingot puller apparatus **100** for growing and pulling an ingot **113** (FIG. 4) from the melt **104**. Pulling mechanism **114** includes a pulling cable **118**, a seed holder or chuck **120** coupled to one end of the pulling cable **118**, and a seed crystal **122** coupled to the seed holder or chuck **120** for initiating crystal growth. One end of the pulling cable **118** is connected to a pulley (not shown) or a drum (not shown), or any other suitable type of lifting mechanism, for example, a shaft, and the other end is connected to the chuck **120** that holds the seed crystal **122**. In operation, the seed crystal **122** is lowered to contact the surface of the melt **104** (FIG. 3). The pulling mechanism **114** is operated to cause the seed crystal **122** to rise. This causes a single crystal ingot **113** (FIG. 4) to be pulled from the melt **104**.

(19) During heating and crystal pulling, a crucible drive unit **107** (e.g., a motor) rotates the crucible assembly **102** and susceptor **106**. A lift mechanism **112** raises and lowers the crucible assembly **102** along the pull axis A during the growth process. For example, the crucible assembly **102** may be at a lowest position (near the bottom heater **126**) in which a charge of solid-phase polycrystalline silicon **133** previously added to the crucible assembly **102** is melted. Crystal growth commences by contacting the melt **104** (FIG. 2) with the seed crystal **122** and lifting the seed crystal **122** by the pulling mechanism **114**.

(20) A crystal drive unit (not shown) may also rotate the pulling cable **118** and ingot **113** (FIG. 4) in a direction opposite the direction in which the crucible drive unit **107** rotates the crucible assembly

102 (e.g., counter-rotation). In embodiments using iso-rotation, the crystal drive unit may rotate the pulling cable **118** in the same direction in which crucible drive unit **107** rotates the crucible assembly **102**.

(21) According to the Czochralski single crystal growth process, a quantity of solid-phase silicon **133** (FIG. 1) such as polycrystalline silicon, or polysilicon, is charged to the crucible assembly **102**. The semiconductor or solar-grade material that is introduced into the crucible assembly **102** is melted by heat provided from one or more heating elements. The size of the solid-phase silicon charge **133** may correspond to the desired size of the melt when ingot growth commences or, as in other embodiments, a smaller charge is used and additional silicon is added by a silicon feed system **200** to form the initial volume of melt present during initiation of ingot growth.

(22) The ingot puller apparatus **100** includes bottom insulation **110** and side insulation **124** to retain heat in the puller apparatus **100**. In the illustrated embodiment, the ingot puller apparatus **100** includes a bottom heater **126** disposed below the crucible floor **129**. The crucible assembly **102** may be moved to be in relatively close proximity to the bottom heater **126** to melt the polycrystalline charged to the crucible assembly **102**.

(23) The silicon melt **104** is stabilized in a stabilization phase previous to the ingot grown phase. In some embodiments, the stabilization phase includes at least the period between formation of the melt (after meltdown) and contact of the surface of the melt with the seed crystal. In some embodiments, the stabilization phase also includes the period at which silicon is melted down in the crucible assembly **102** (i.e., meltdown of the initial charge and any other silicon added to form the initial melt).

(24) To form the ingot, after the melt is stabilized in the stabilization phase, the seed crystal **122** is contacted with the surface **111** of the melt **104** as shown in FIG. 3. The pulling mechanism **114** is operated to pull the seed crystal **122** from the melt **104** in an ingot growth phase. Referring now to FIG. 4, the ingot **113** includes a crown portion **142** in which the ingot transitions and tapers outward from the seed crystal **122** to reach a target diameter. The ingot **113** includes a constant diameter portion **145** or cylindrical “main body” of the crystal which is grown by increasing the pull rate. The main body **145** of the ingot **113** has a relatively constant diameter. The ingot **113** includes a tail or end-cone (not shown) in which the ingot tapers in diameter after the main body **145**. When the diameter becomes small enough, the ingot **113** is then separated from the melt **104**.

(25) The ingot puller apparatus **100** includes a side heater **135** and a susceptor **106** that encircles the crucible assembly **102** to maintain the temperature of the melt **104** during crystal growth. The side heater **135** is disposed radially outward to the crucible sidewall **131** as the crucible assembly **102** travels up and down the pull axis A. The side heater **135** and bottom heater **126** may be any type of heater that allows the side heater **135** and bottom heater **126** to operate as described herein. In some embodiments, the heaters **135**, **126** are resistance heaters. The side heater **135** and bottom heater **126** may be controlled by a control system (not shown) so that the temperature of the melt **104** is controlled throughout the pulling process.

(26) The ingot puller apparatus **100** may include a heat shield **151**. The heat shield **151** may shroud the ingot **113** and may be disposed within the crucible assembly **102** during crystal growth (FIG. 4). The ingot puller apparatus **100** may include an inert gas system to introduce and withdraw an inert gas such as argon from the growth chamber **152**.

(27) The ingot puller apparatus **100** also includes a silicon feed system **200** for introducing solid-state silicon (e.g., polycrystalline silicon or single crystal silicon scrap material) into the crucible assembly **102** and, in particular, the outer melt zone **143**. The solid-state silicon may be added continuously during ingot growth to maintain a substantially constant melt elevation level and volume during growth of the ingot **113** or may be added intermittently between crystals. The solid silicon that is fed to the crucible assembly **102** by the silicon feed system **200** may be, for example, granular, chunk, chip, or a combination of thereof. The silicon feed system **200** adds or meters solid-state silicon into and through a silicon feed tube **214**. Referring now to FIG. 5, an example

silicon feed system **200** of the present disclosure is shown. The silicon feed system **200** includes a canister **204** for holding solid-state silicon. The canister **204** includes a removable lid **207** for sealing the canister **204**. The canister **204** also includes a funnel portion **211** which directs silicon to a feed tray **260** disposed below the canister **204**.

(28) To move silicon from the canister **204** to the feed tube **214**, the silicon feed system **200** includes a feed tray **260** disposed below the canister **204** that extends between the canister **204** and a port **229** (FIG. **6**) that extends through the ingot puller housing **108**. The feed tray **260** is disposed above and is connected to a vibrator **262**. Powering the vibrator **262** causes the feed tray **260** to vibrate which conveys silicon from the canister **204** and across the feed tray **260**. Silicon is discharged from the feed tray **260** and into a port opening **232** (FIG. **6**) (e.g., through a tube or chute (not shown) disposed below the feed tray **260**). Silicon falls through the port opening **232** and into a funnel **222** disposed below the port **229**. From the funnel **222** the silicon falls into the silicon tube **214**. Silicon may be added to maintain a constant position of the melt surface and may be controlled by the power supplied to the vibrator **262**.

(29) The port **229** and silicon feed tube **214** are sealed with the canister **204** to isolate the growth chamber **152** (FIG. **1**) of the ingot puller apparatus **100** from the surrounding ambient. For example, the silicon feed system **200** may include an outer housing **252** with other components of the feed system being disposed within the outer housing **252**. An inert gas such as argon may be introduced from an inert gas feed vessel **255** into the outer housing **252**. The inert gas passes through the port opening **232** (FIG. **6**) and into the silicon feed tube **214**.

(30) The silicon feed system **200** is exemplary and any feed system that enables silicon to be introduced into the crucible assembly **102** may be used. For example, other feed systems may include liquid feed systems.

(31) Referring now to FIG. **6**, the tube **214** includes a receiving portion **225** and a main portion **227** disposed below the receiving portion **225**. The funnel **222** is partially disposed within the receiving portion **225** of the silicon feed tube **214**. The receiving portion **225** of the tube includes a constant diameter segment **231** and a tapering segment **223** that tapers to the main portion **227** of the tube **214**. The main portion **227** of the tube **214** has an angled segment **243** (i.e., a segment that extends radially inward in the puller apparatus **100**) and a terminal segment **245** that extends from the angled segment **243**. The silicon tube **214** includes an upper end **221** and a lower end **249**.

(32) In accordance with embodiments of the present disclosure, the amount of inert gas that is introduced into the port **229** and that travels through the funnel **222** and the silicon tube **214** is regulated to reduce downstream clogging of the tube **214**. For example, different flow rates may be used during the stabilization phase and the ingot growth phase. During the stabilization phase, the inert gas is introduced into the silicon feed tube at a first inert gas feed rate. During the ingot growth phase, the inert gas is introduced into the silicon feed tube at a second inert gas feed rate. The first inert gas feed rate is greater than the second inert gas feed rate. For example, the ratio of the first inert gas feed rate to the second inert gas feed rate may be at least 5:4, at least 4:3, at least 3:2 or at least 2:1. In some embodiments, the first inert gas feed rate is greater than 3 slpm and the second inert gas feed rate is from 1 slpm to 3 slpm. For example and in accordance with some particular embodiments, the first inert gas feed rate and the second inert gas feed rate are each between 2.5 and 3.5 slpm.

(33) It should be noted that the inert gas feed rates and ratios described above may apply to the entire respective stabilization phase or inert gas phase or, as in some embodiments, at least 75% of the respective phase, or at least 85%, at least 95%, or at least 99% of the respective phase.

(34) In some embodiments, there may be a gap **233** (e.g. open gap as shown in FIG. **6**) between the funnel **222** and the silicon feed tube **214**. Some inert gas may exit the gap **233** instead of proceeding through the length of the tube **214**. The flow rates and/or ratios of flow rates described herein may refer to the amount of gas introduced through the port opening **232** (i.e., upstream of any gases passing through gap **233**).

(35) Referring now to FIGS. 7-8, in some embodiments, the silicon feed tube **214** and/or funnel **222** may be modified relative to conventional tubes and funnels. The feed tube **214** and/or funnel may reduce clogging of the tube **214** relative to conventional tubes and funnels. The silicon feed tube **214** and/or funnel **222** described below may be used in combination with the inert gas flow rates and ratios described above or the flow rates and ratios may be used without the modified tube **214** and funnel **222** and the modified tube **214** and funnel **222** may be used with different inert gas flow rates and ratios.

(36) As shown in FIG. 7, the funnel **222** at least partially extends into the silicon feed tube **214**. The funnel **222** includes a tapering portion **237** that is disposed below the opening **232** of the port **229**. The funnel **222** has a chute portion **240** that partially extends into the silicon feed tube **214**. The funnel **222** has an upper end **212** and a lower end **215** with the lower end **215** being disposed within the silicon feed tube **214**. The silicon feed tube **214** includes a receiving portion **225** in which the chute portion **240** of the funnel **222** is partially disposed. The receiving portion **225** has a diameter larger than the main portion **227** of the tube **214**.

(37) The funnel **222** and silicon feed tube **214** are separated by a gap **219**. The ingot puller apparatus **100** includes a flange **230** that extends radially from the funnel **222**. The flange **230** extends across at least a portion of the gap **219** (e.g., the entire gap **219** as shown in FIG. 7 or partially across the gap **219** as shown in FIG. 8) to reduce backflow of gases from the silicon feed tube **214** into the growth chamber **152** (FIG. 1).

(38) In the embodiment illustrated in FIG. 7, the flange **230** extends from the funnel **222** and is disposed above the silicon feed tube **214** (i.e., directly above). The flange **230** contacts an upper end **221** of the feed tube **214**. In the embodiment illustrated in FIG. 8, the flange **230** extends from the funnel **222** and is disposed within the silicon feed tube **214** (e.g., the receiving portion **225** of the tube **214**). The flange **230** only partially extends across a portion of the gap **219**. In the illustrated embodiment, the flange **230** extends from the lower end **215** of the funnel **222**. In some embodiments, the ingot puller apparatus of FIG. 7 is modified to also include the flange shown in FIG. 8 (i.e., the ingot puller apparatus **100** includes both flanges). The flange **230** of FIG. 7 and the flange **230** of FIG. 8 may extend from either the feed tube **214** or funnel **222**.

(39) The flange **230** may be made of quartz (e.g., devitrified quartz) or silicon carbide. The flange **230** may be a separate ring that is connected to the tube **214** or funnel **222** or may be integral with the tube **214** or funnel **222** (e.g., as one manufactured part).

(40) It should be noted that in embodiments in which the ingot puller apparatus has been modified from conventional apparatus, generally any ingot puller apparatus that includes a flange that extends across a portion of the gap to reduce backflow of gases from the silicon feed tube into the growth chamber may be used unless stated otherwise. The ingot puller apparatus shown in FIGS. 7-8 are example apparatus and other arrangements of silicon tubes, funnels and/or flanges may be used. While the silicon feed tube **214** has been described with reference to addition of silicon, the tube **214** may be used to add dopant or quartz cullets to the silicon melt.

(41) Compared to conventional methods, the methods of the present disclosure for growing a single crystal silicon ingot in a continuous Czochralski process have several advantages. Without being bound by any particular theory, it is believed that clogging of the feed tube is caused by chemical vapor deposition of silicon oxides ($\text{SiO}_{x\text{sub}}$) on the inner surface of the feed tube. The silicon oxide originates from the silicon melt from which it evaporates and rises inside the feeding tube due to buoyancy forces. Once inside the feed tube, silicon oxide rises and encounters lower temperature regions causing deposition of metallic silicon and silicon dioxide by the formula:

$$2\text{SiO}_{x\text{sub}} \rightarrow \text{SiO}_{2\text{sub}} + \text{Si} \quad (1).$$

(42) It is believed that by reducing the SiO concentration to 1 wt % or less, clogging in the feed tube may be prevented. During the stabilization phase, it is more difficult to suppress the concentration of SiO due to the higher temperatures used to keep the silicon molten. Clogging of the feed tube causes the ingot production run to be stopped and the ingot puller apparatus to be

taken out of service. By controlling the flow rate of inert gas that passes through the silicon feed tube such that the flow rate is higher during the stabilization phase and relatively lower during the ingot growth phase, the flow rate is increased during high temperature stabilization phase (stabilization contributing to more clogging than the ingot growth phase due to its higher temperature) and the flow rate is decreased during ingot growth which reduces inert gas bubbles which cause defects in the ingot.

(43) In embodiments in which the ingot puller apparatus includes a flange that extends radially from the funnel or from the silicon feed tube and extends across at least a portion of a gap between the funnel and the feed tube, the flange reduces backflow of gases from the silicon feed tube into the growth chamber.

(44) In embodiments in which the inert gas feed rate is near 3 slpm (e.g., 2.5 slpm to 3.5 slpm) in both the stabilization phase and ingot growth phase, the concentration of SiO in the silicon feed tube may be maintained at 1.0 wt % at all stages of the ingot growth process which is believed to be the threshold at which the CVD reaction occurs.

EXAMPLES

(45) The processes of the present disclosure are further illustrated by the following Examples. These Examples should not be viewed in a limiting sense.

Example 1: SiO Mass Fraction in the Silicon Feed Tube During the Stabilization Phase and the Ingot Growth Phase

(46) FIG. 9 is a simulation of the mass fraction of silicon oxide (SiO) during the ingot growth phase and FIG. 10 is a simulation of the mass fraction of SiO during the stabilization phase. During the ingot growth phase (FIG. 9), the flow of SiO into the silicon feed tube was 0.0132 g/s. During the stabilization phase (FIG. 10), the flow of SiO into the silicon feed tube was 0.0142 g/s. As can be seen from the Figures, the stabilization phase included a higher mass fraction of SiO compared to the ingot growth phase. As shown in FIG. 10, the concentration of SiO was higher at the junction between the angled segment of the tube and the terminal segment of the tube (i.e., at the lower bend in the tube) where clogging most often occurs. During stabilization, the temperature at this junction is between 900° C. and 1000° C. which is within the critical temperature range for the CVD reaction.

Example 2: Reduction in Wafer Defects by Reducing Inert Gas Flow Rate

(47) Three single crystal silicon ingots were prepared by continuous Czochralski with an in ingot puller apparatus shown in FIG. 1. Crystal 1 and Crystal 2 were grown using an argon flow rate of 12 slpm through the silicon feed tube 214 and Crystal 3 was grown using a flow rate of 3 slpm. The count of the defects on the surface of wafers sliced from the ingots was determined for each ingot and is shown in FIG. 11. As shown in FIG. 11, the ingot grown with an argon flow rate of 3 slpm through the silicon feed tube 214 had significantly less defects.

(48) As used herein, the terms “about,” “substantially,” “essentially” and “approximately” when used in conjunction with ranges of dimensions, concentrations, temperatures or other physical or chemical properties or characteristics is meant to cover variations that may exist in the upper and/or lower limits of the ranges of the properties or characteristics, including, for example, variations resulting from rounding, measurement methodology or other statistical variation.

(49) When introducing elements of the present disclosure or the embodiment(s) thereof, the articles “a,” “an,” “the,” and “said” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” “containing,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. The use of terms indicating a particular orientation (e.g., “top,” “bottom,” “side,” etc.) is for convenience of description and does not require any particular orientation of the item described.

(50) As various changes could be made in the above constructions and methods without departing from the scope of the disclosure, it is intended that all matter contained in the above description and

shown in the accompanying drawing[s] shall be interpreted as illustrative and not in a limiting sense.

Claims

1. An ingot puller apparatus for manufacturing a single crystal silicon ingot, the ingot puller apparatus comprising: a crucible assembly for holding a silicon melt; a crystal puller housing that defines a growth chamber for pulling a silicon ingot from the melt, the crucible assembly being disposed within the growth chamber; a silicon feed tube for adding solid silicon to the crucible assembly, the silicon feed tube extending into the growth chamber; a funnel that at least partially extends into the silicon feed tube, the funnel and silicon feed tube being separated by a gap; and a flange that extends from the silicon feed tube, the flange extending fully across the gap to reduce backflow of gases from the silicon feed tube into the growth chamber, the flange being disposed above the silicon feed tube and contacting an upper end of the silicon feed tube, the funnel extending through the flange.
 2. The ingot puller apparatus as set forth in claim 1 wherein the flange is a quartz ring.
 3. The ingot puller apparatus as set forth in claim 1 wherein the silicon feed tube comprises a receiving portion and a main portion having a diameter less than the receiving portion, the funnel being at least partially disposed within the receiving portion.
 4. The ingot puller apparatus as set forth in claim 1 wherein the flange and crystal puller housing are separated by a gap.
 5. An ingot puller apparatus for manufacturing a single crystal silicon ingot, the ingot puller apparatus comprising: a crucible assembly for holding a silicon melt; a crystal puller housing that defines a growth chamber for pulling a silicon ingot from the melt, the crucible assembly being disposed within the growth chamber; a silicon feed tube for adding solid silicon to the crucible assembly, the silicon feed tube extending into the growth chamber; a funnel that at least partially extends into the silicon feed tube, the funnel and silicon feed tube being separated by a gap; and a flange that extends from the silicon feed tube, the flange extending fully across the gap to reduce backflow of gases from the silicon feed tube into the growth chamber, the flange being disposed within the silicon feed tube, the flange having an opening through which solid silicon passes.
 6. The ingot puller apparatus as set forth in claim 5 wherein the flange is made of quartz.
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